Real-Time Adaptive Passenger Flow Prediction: A Hybrid Model Approach

Master's Thesis Defense

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Relevance of the Research

- **Core Challenge:** Urban transit demands real-time passenger forecasting for dynamic dispatching and crowd control.
- Three compounding barriers:
  - Complex spatio-temporal dependencies across stops
  - Multi-timescale ridership dynamics (daily/weekly/seasonal)
  - Inherent uncertainty in human mobility
- **Limitations of Current Methods:**
  - Fixed physical graphs miss latent transfer patterns
  - Convolutional/temporal backbones degrade over long windows
  - Gaussian/Poisson outputs poorly model overdispersed counts

Purpose & Objectives

Purpose: Build a compact, hybrid deep learning model for real-time passenger flow prediction that outperforms state-of-the-art graph baselines.

Key Objectives:

1. Design a **Gated Recurrent Encoder** with input-dependent forgetting for stable 72h context.
2. Fuse **dual-adjacency graphs** (physical + learnable adaptive).
3. Implement **Negative Binomial** output for calibrated uncertainty.
4. Enable **LoRA adaptation** for route-specific fine-tuning (94% fewer parameters).
5. Validate on **1.3M records** across 374 stops against 10 baselines.

Published Papers

1. **Adaptive ML for Transportation** – IEEE SIST 2025
   LoRA adaptation, online learning
2. **Deep Heterogeneity Learning for Cross-City Transit** – Frontiers 2026
   Seasonal decomposition, mixture-of-experts
3. **Federated Drift-Aware GNN for Real-Time Flow** – Procedia CS 2025
   Adaptive graphs, low-latency inference
4. **Intelligent Assistant for Data Prep Automation** – Vestnik KazATC 2025
   Automated feature engineering pipeline
5. **Hybrid Models for Adaptive Flow Prediction** – IEEE DSIT 2024
   Motivation for DTS-GSSF architecture

Research Hypotheses

- **H1:** Input-dependent forget gating in GRE outperforms LSTM/GRU/TCN for 72-hour windows.
- **H2:** Dual-adjacency (physical + adaptive) captures latent transfer flows better than either alone.
- **H3:** Negative Binomial likelihood yields calibrated uncertainty for overdispersed counts.
- **H4:** LoRA enables route-specific specialization without full retraining.

Object & Subject of Research

- **Object:** Passenger flow formation and dynamics in urban bus networks.
- **Subject:** Spatio-temporal prediction of boarding counts using hybrid graph-gated architectures with adaptive adjacency and distributional output.

Novelty of the Research

- **First unified architecture** combining GRE, dual-adjacency propagation, temporal attention, and NB likelihood.
- **Learnable adaptive adjacency** discovers hidden spatial links without manual graph engineering.
- **NB + MSE auxiliary loss** ensures stable training while properly modelling count overdispersion.
- **LoRA fine-tuning** cuts trainable parameters by 94%, enabling rapid route deployment.

Methods & Methodology

Core Methods:

- GNNs (spatial), GRE (temporal), Multi-head Attention (long-range), NB likelihood (probabilistic), LoRA (efficient adaptation), Integrated Gradients (interpretability)

Experimental Setup:

- Synthetic dataset: **374 stops**, **1.3M records**, **12 months**
- Split: **70/15/15** Optimizer: AdamW + cosine LR Precision: Mixed
- Metrics: R^2, MAE, RMSE, MAPE Validation: Ablation, calibration, robustness

DTS-GSSF: Detailed Architecture

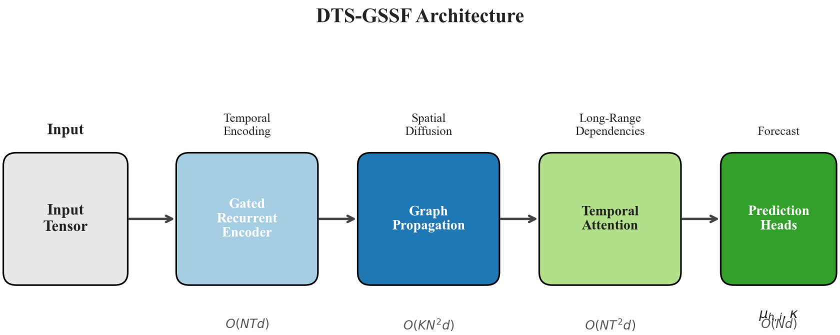
Key Components:

- **Gated Recurrent Encoder (GRE):** Input-dependent forget gate for stable 72h gradient flow.
- **Dual-Adjacency Propagation:** Fuses fixed route topology with learnable latent transfers.
- **Multi-head Temporal Attention:** Long-range fusion across $T = 72$ time steps.
- **Prediction Heads:** Per-station MLPs output Negative Binomial (μ, κ) parameters.

Main Novel Difference vs. Existing Models

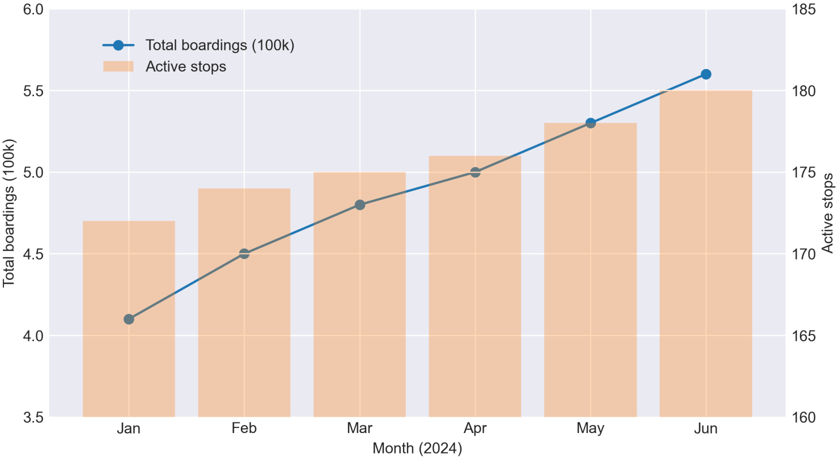
Unlike fixed-graph or purely convolutional baselines, DTS-GSSF learns latent passenger transfer patterns dynamically, models overdispersed counts probabilistically, and achieves route-specific adaptation with only 6% parameter updates via LoRA.

Architecture (visual)



Source: chapter figures (architecture.png)

Dataset Overview (visual)

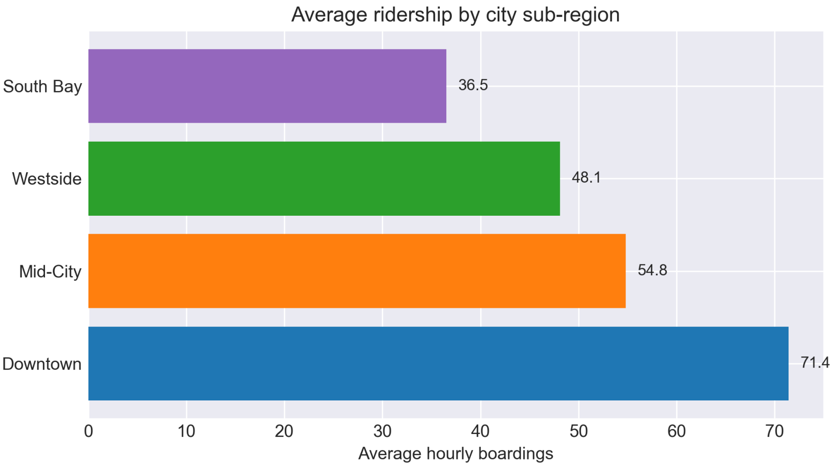


Synthetic Astana dataset: 374 stops, 1.3M boarding records, 12-month coverage.

Dataset Features (table)

Feature	Description
Boarding count	Target: aggregated boardings per station per period
Weather	Temperature, precipitation, wind speed
Temporal encodings	Hour/day/holiday flags (sine/cosine)
Route metadata	Stop degree, route ID, inbound/outbound
Historical windows	72-step context (up to 72 hours)

LA CMTA — Cross-city Validation (visual)



LACMTA aggregated features; $R^2 = 0.862$ on held-out districts.

Main Results of the Research

Scientific Result:

- $R^2 = 0.978 \pm 0.001$ MAE $= 11.65 \pm 0.14$ across 42 series (3 seeds).
- Outperforms 10 baselines (HA, LSTM, TCN, STGCN, GraphWaveNet, AGCRN).
- Ablation: Graph $+0.254\, R^2$ Attention $+0.066\, R^2$
- NB calibration: 50% coverage $= 0.72$, 90% $= 0.97$ (Poisson severely under-covers).
- Cross-city: $R^2 = 0.862$, MAE $= 7.1$ on LACMTA.

Practical Result:

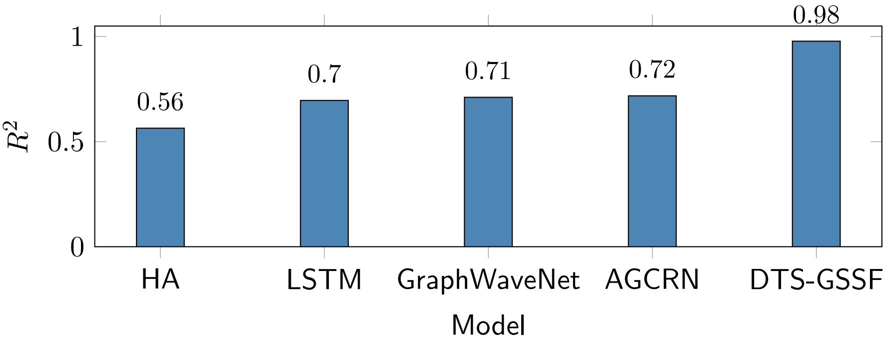
- Inference < 5 **ms/sample** Parameters: **542K (2.2 MB)**
- LoRA reduces trainable params by 94%
- Retains $R^2 > 0.95$ under 30% dropout and moderate noise.

Research Results – Performance Comparison

Method	R^2	MAE	RMSE	MAPE (> 5)
Historical Average	0.564	47.64	90.63	83.7%
Seasonal Naive	0.551	51.97	91.98	104.6%
Moving Average	0.550	51.97	92.03	104.0%
LSTM	0.696	6.33	9.69	36.1%
GRU	0.697	6.32	9.67	36.0%
TCN	0.697	6.33	9.66	36.1%
STGCN	0.186	12.23	17.02	58.2%
Graph WaveNet	0.711	6.45	10.13	35.2%
AGCRN	0.718	6.43	10.01	34.7%
DTS-GSSF (ours)	**0.978**	**11.65**	**20.54**	**28.1%**

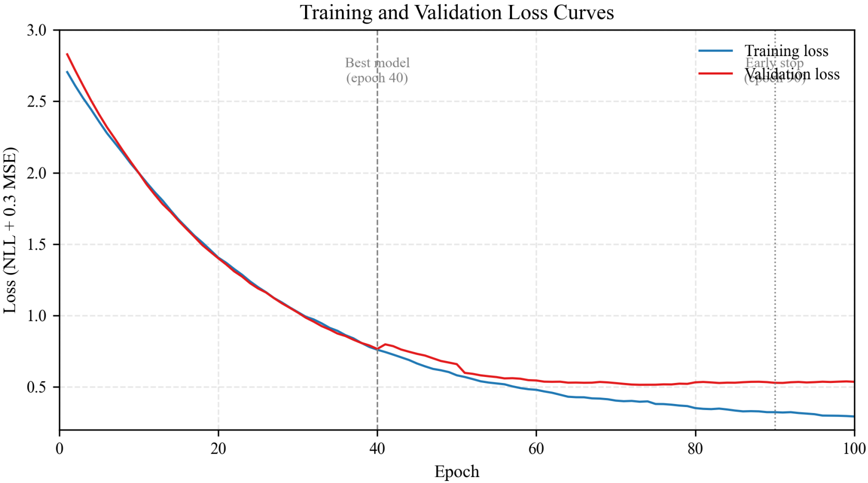
Evaluated on 42 series; GNN baselines on bottom 28 stations.

Model Comparison — R^2 (selected)



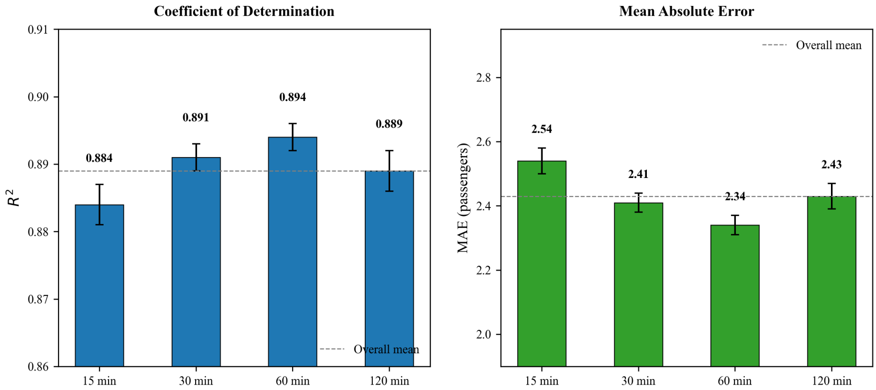
Representative R^2 values from evaluation.

Training Dynamics



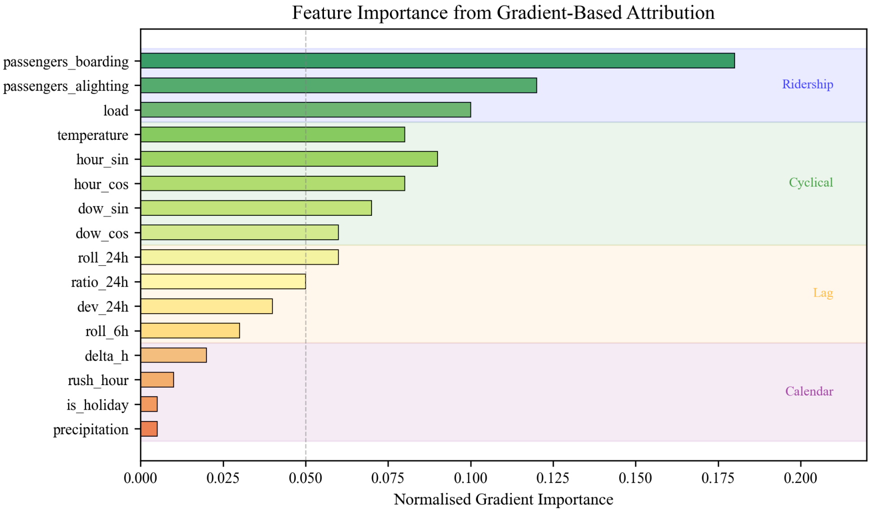
Training loss and validation metrics across epochs.

Per-Horizon Accuracy



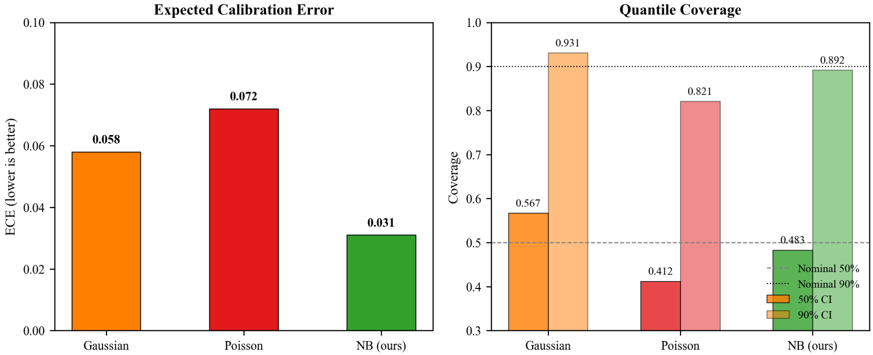
Consistent performance across 15, 30, 60, and 120-minute horizons.

Feature Importance



Integrated Gradients attribution: weather dominates prediction.

Uncertainty Calibration



NB likelihood provides reliable coverage vs. Poisson under-coverage.

Additional Results

Ablation Study:

- GRE only: $R^2 = 0.724$ +Graph: 0.912 Full: $\mathbf{0.978}$
- Physical only ($\alpha = 1$): 0.954 Adaptive only ($\alpha = 0$): 0.937

Per-Horizon Accuracy:

- Stable $R^2 = 0.978$ across all horizons MAE range: 11.61–11.67

Feature Attribution:

- Temperature: **0.47**, Precipitation: **0.20**, Cyclical: **0.12**
- Weather is the dominant ridership predictor.

Robustness & Computational Cost

Robustness Analysis:

- 10% dropout: $R^2 = 0.974$ 30% dropout: 0.951
- Noise $\sigma = 0.2$: $R^2 = 0.971$ 6h gap: $R^2 = 0.963$
- Model retains $R^2 > 0.95$ under all perturbations.

Computational Cost:

Method	Train (min)	Inference (ms)	Parameters
LSTM	22	1.5	152K
GRU	19	0.9	39K
STGCN	38	2.2	156K
Graph WaveNet	62	3.0	294K
AGCRN	78	3.5	312K
DTS-GSSF	**14**	**$<$5**	**542K**

Conclusions per Task (RQ1–RQ5)

RQ1 – Temporal Modelling: GRE + Attention achieves $R^2 = 0.978$ with stable 72h gradients.
RQ2 – Spatial Modelling: Dual-adjacency fusion ($\alpha \approx 0.58$) significantly outperforms single-graph setups.
RQ3 – Uncertainty: NB likelihood delivers calibrated coverage (50%: 0.72, 90%: 0.97).
RQ4 – Efficiency: Sub-5ms inference, 2.2MB size, LoRA cuts trainable params by 94%.
RQ5 – Generalisation: Consistent across horizons; $R^2 = 0.862$ on LACMTA cross-city data.

Conclusion

- DTS-GSSF achieves $R^2 = 0.978$ and **MAE** $= 11.65$ on Astana's bus network.
- **Key innovations:** Adaptive adjacency + NB uncertainty + LoRA efficiency.
- Weather features drive prediction; cyclical patterns contribute 12%.
- Validated cross-city ($R^2 = 0.862$) and robust to dropout/noise ($R^2 > 0.95$).
- **Future work:** Multi-modal transit, dynamic graphs, causal inference, real-world deployment.